

Input Form (IF)

Score: 3/5 — Moderate gaps | Confidence: high

Benchmark: MILU | Context: Mymensingh Environmental Scientist — Cross-Regional South Asian Agricultural Knowledge

Input Form definition: Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

MILU's text-based MCQ format is technically compatible with the deployment's text-only modality, and Bengali and Telugu scripts are reliably rendered (DATASET strengths 1, 4). However, two concerns reduce alignment: (a) the exam-register MCQ phrasing differs from naturalistic professional-query style (elicitation A4), and (b) approximately 25% of questions are machine-translated [Q43], with the sampled Bengali validation split observed at 100% translated (DATASET-D3, CRITICAL Concern 2), introducing register artifacts and occasional pipeline quality issues including incomplete question stems (DATASET-D34, D35, D36 — MINOR Concern 8). For a MODERATE-priority dimension where script support is the floor requirement, MILU clears that bar but does not match the naturalistic register a Mymensingh domain expert would use.

ID	Evidence
Q43	"Table 2 shows the overall statistics of MILU. Of the total 79K questions, only 25% of questions are translated from English, with the remainder" (p.4)
D3	**is_translated:** True — All 21 Bengali validation examples are machine-translated from English
D34	"নচিৰে প্ৰশ্ন... সেই বকল্‌পট নিৰ্বাচন কৰুন যা প্ৰদত্ত শব্দৰে বপীৰীত অৰ্থ প্ৰকাশ কৰে" (question body missing the target word) — Antonym question — incomplete stem, illustrates quality issues in translated questions
D35	"নচিৰে বৰিতিটি দিউ যুক্তি এবং II সহ দেওয়া হয়ছে" (arguments not shown in question field) — Argument evaluation question — missing argument content, shows data quality gap in translated items
D36	The owner of the textile shop brought a... Calculator — Incomplete question stem — no meaningful question context present

Strengths

- Bengali and Telugu scripts are fully populated in 215 sampled examples (DATASET Strength 1: D3, D31), confirming script-level encoding compatibility for the deployment's primary languages
- Multiple cleaning layers were applied — INDICLID language ID, Unicode filtering, deduplication, manual spot-checks [Q30, Q31, Q32] — providing baseline data hygiene
- Consistent 4-option MCQ structure across all 11 languages enables reproducible cross-language and cross-model comparison (DATASET Strength 4: D6, D11)

ID	Evidence
Q30	"To remove incorrect language entries, we utilize a combination of INDICLID (Madhani et al., 2023) and Unicode-based filtering (Khan et al., 2024), ensuring that the questions are in the correct language." (p.4)
Q31	"To further refine the dataset, we remove any duplicate questions to retain only the unique ones." (p.4)
Q32	"As a final step, we manually verify a sample of questions from each language to ensure accuracy and correct any remaining errors." (p.4)
D3	**is_translated:** True — All 21 Bengali validation examples are machine-translated from English
D6	A geographic region has the following distinct characteristics: 1. Warm and dry climate 2. Mild and wet winter 3. Evergreen oak trees — Mediterranean climate region identification — generic geography
D11	ರಾಷ್ಟ್ರೀಯ ಆಹಾರ ಭದ್ರತೆ ಮಂಷನ್ ಒಂದು ಬೆಳೆ ಅಭಿವೃದ್ಧಿ ಯೋಜನೆ... ಮಣ್ಣಿನ ಆರೋಗ್ಯವನ್ನು ಪುನಃಸ್ಥಾಪಿಸಲು — National Food Security Mission crop scheme in Kannada — same Indian policy content replicated across languages

D31	"স্বথরি প্রবাহ ট্রান্সফরমার _____ ধরনের। ... শলে" (is_translated: True) — Shell-type constant current transformer — engineering question, machine-translated from English
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Checklist

Checklist item definitions:

ID	Assessment Question
IF-1	Compare signal distributions between source and target
IF-2	Check if regional infrastructure supports data capture specs
IF-3	Identify domain-specific form differences
IF-4	Document form mismatches harming external validity

Checklist responses:

IF-1: The text-based signal modality matches deployment needs (both are text). However, register distribution differs: MILU is exam-register MCQ, deployment is naturalistic professional query (elicitation Q4/A4). Script encoding in Bengali and Telugu is verified functional [DATASET-D3, D31].

IF-2: Mymensingh has connectivity adequate for text-based LLM access — Bangladesh has ~47% internet penetration with ~100% 4G coverage as of 2025 [WEB-20, WEB-21], and Mymensingh hosts BAU and is a divisional capital [WEB-7]. Text MCQ infrastructure is supported.

IF-3: Domain-specific form differences: (a) machine-translated content carries register artifacts — the YAML notes 'translated questions likely carry Indian Bengali register conventions rather than Bangladeshi Bengali norms', and (b) some translated items have incomplete stems [DATASET-D34, D35] suggesting pipeline degradation. MCQ format also excludes reading-comprehension and image-based queries [Q29], though this is non-binding for the deployment.

IF-4: Form mismatches are moderate: register artifacts from machine translation [Q43, Q84] (DATASET-D3 — 100% translation in Bengali validation), occasional truncated question stems [DATASET-D34, D35, D36], and exam-register vs. naturalistic-query phrasing divergence. None block use, but they reduce ecological validity.

ID	Evidence
Q4	"Open multilingual models outperform language-specific fine-tuned models, which perform only slightly better than random baselines." (p.1)
Q29	"During the collection process, we exclude any reading-comprehension-style questions, images-based questions, and those with more than four answer options to ensure uniformity and consistency." (p.4)
Q43	"Table 2 shows the overall statistics of MILU. Of the total 79K questions, only 25% of questions are translated from English, with the remainder" (p.4)
Q84	"Third, the scarcity of questions necessitated translating a portion of the dataset." (p.9)
D3	**is_translated:** True — All 21 Bengali validation examples are machine-translated from English
D31	"স্বথরি প্রবাহ ট্রান্সফরমার _____ ধরনের। ... শলে" (is_translated: True) — Shell-type constant current transformer — engineering question, machine-translated from English
D34	"নচিরে প্রশ্ন... সেই বকল্‌পটিনিবিচান কব্বুন যা প্রদত্ত শব্দরে বপিরীত অর্থ প্রকাশ করে" (question body missing the target word) — Antonym question — incomplete stem, illustrates quality issues in translated questions
D35	"নচিরে বব্বিতটিদুটযুক্তি এবং II সহ দেওয়া হয়ছে" (arguments not shown in question field) — Argument evaluation question — missing argument content, shows data quality gap in translated items
D36	The owner of the textile shop brought a... Calculator — Incomplete question stem — no meaningful question context present
WEB-7	BAU as professional institutional context — text infrastructure supported (https://bau.edu.bd/)
WEB-20	Bangladesh internet/mobile coverage adequate for text modality (https://datareportal.com/reports/digital-2024-bangladesh)
WEB-21	Mobile-dominant connectivity in Bangladesh (https://en.wikipedia.org/wiki/Internet_in_Bangladesh)

Information Gaps

- Whether the test split (vs validation) has a lower machine-translation rate for Bengali
- Mymensingh-specific connectivity statistics (national aggregates only [WEB-20])

Requires Expert Verification

- A Bangladeshi Bengali linguist's review of GPT-4O-translated MILU items for register naturalness vs. Indian Bengali bias

Remediation

Gap 1: Exam-register MCQ phrasing diverges from naturalistic professional-query register; some translated items have truncated stems

Recommendation: Augment evaluation with naturalistic-register Bangladeshi Bengali agricultural query items elicited from BAU/BARI scientists; filter out items with incomplete stems [DATASET-D34, D35, D36] before reporting scores.

ID	Evidence
D34	"নচিৰে প্ৰশ্ন... সেই বকিল্পট নিৰ্বাচন কৰুন যা প্ৰদত্ত শব্দৰে বপীৰীত অৰ্থ প্ৰকাশ কৰে" (question body missing the target word) — Antonym question — incomplete stem, illustrates quality issues in translated questions
D35	"নচিৰে ববিতটি দুটি যুক্তি এবং II সহ দেওয়া হয়েছে" (arguments not shown in question field) — Argument evaluation question — missing argument content, shows data quality gap in translated items
D36	The owner of the textile shop brought a... Calculator — Incomplete question stem — no meaningful question context present